# CIS 6270: Discrete Generative Models

Fall 2026

## Instructor

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## TAs

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## Class Times and Location

Tuesdays and Thursdays, 8:30 - 10 am in Towne 337

## Office Hours Location

Scheduled per session at either: One uCity Square Lobby or Amy Guttman Hall

## Course Description

Generative models operate on continuous and discrete state spaces. This course develops their mathematical, algorithmic, and empirical foundations, with discrete state modeling as the main focus. We begin with probability, linear algebra, differential equations, information theory, maps, pushforward distributions, changes of variables, and conservation of probability. We then develop continuous generative models through ordinary and stochastic differential equations, flow matching, diffusion, score matching, denoising objectives, probability-flow ODEs, and guidance. These constructions lead to discrete diffusion, including masked and uniform corruption, categorical posteriors, continuous-time jump processes, block diffusion, and guided

sequence generation. We next derive discrete flow matching through probability paths, target rate matrices, conditional and marginal rates, simplex-valued interpolants, Fisher-Rao geometry, Gumbel-softmax flows, rectification, and multi-objective guidance. The course then studies one-step and few-step generation through flow maps, consistency objectives, MeanFlow, discrete and latent flow maps, posterior guidance, expanding state spaces, and stochastic flow maps. The final lecture connects these methods to optimal transport and Schrödinger bridges through couplings, the Monge and Kantorovich problems, Wasserstein distances, entropic regularization, Sinkhorn iterations, path-space relative entropy, stochastic control, Doob transforms, iterative proportional fitting, and discrete bridge matching. Lectures emphasize derivations of objectives, transition kernels, rate matrices, probability-evolution equations, training losses, and sampling rules, together with worked numerical examples and analyses of computational behavior. Integrated code examples give students experience implementing architectures, training objectives, samplers, guidance procedures, and evaluations. Applications and projects span biological sequences and structures, images, audio, natural language, code, graphs, point clouds, 3D molecular data, probability-simplex representations, and continuous latent representations. By the end of the semester, students will be able to derive and analyze continuous and discrete generative processes, translate mathematical specifications into working algorithms, evaluate models across data modalities, and read and contribute to current research in generative modeling.

## Learning Objectives

1. Build a rigorous understanding of the mathematical foundations of generative models, including probability, the probability simplex, continuous-time Markov chains, the master equation, and generative transport frameworks.

2. Derive, analyze, and implement the core generative frameworks: classical generative models, discrete diffusion, discrete flow matching, optimal transport, stochastic control, and discrete Schrödinger bridges.

3. Apply these frameworks across biological sequences and structures, images, audio, natural language, code, graphs, point clouds, 3D molecular data, probability-simplex representations, and continuous latent representations.

4. Connect derivations to implementations by building algorithms from scratch and studying their behavior empirically through benchmarks and ablations.

5. Read current research and design, communicate, and defend novel discrete generative modeling strategies for sequence design and analysis.

## Prerequisites

Students must have completed one of the following courses: CIS 5190, CIS 5200, or ESE 5460.

## Communications

Course communications will be via Canvas: `https://canvas.upenn.edu/`. Students are also encouraged to email the instructor and TAs.

## Course Website

All lecture notes, code, slides, and course information are available on our course website: `https://cis6270.github.io/`.

## Textbooks

The primary textbook is *Discrete Generative Models*, Professor Chatterjee's complete lecture notes covering all course material, which you can find on the course website.

## Grading

- **Participation (10%):** Active participation in lecture and discussion is expected. This includes contributing to proofs and derivations in class, engaging in algorithm design sessions, and working through implementation examples. Students are encouraged to arrive to class on time each day, but must arrive by 8:40AM to be marked present. *Starting at 8:41AM, new arrivals will be considered absent.* Unexcused absences will reduce the final grade (1 point per absence, 0.5 if reported at least a week in advance). The instructor and TAs reserve the right to deduct points based on level of contribution to the class.

- **Exams (35%):** Two written exams (Exam 1: 15 points; Exam 2: 20 points), closed-book with a single reference sheet allowed. Exams emphasize rigorous mathematical reasoning and algorithm design. Students will construct formal proofs of convergence, invariance, or optimality; derive equations central to discrete generative models (for example forward and reverse kernels in diffusion, the master equation and rate matrices, the generating velocity and continuity equations in flow matching, and variational bounds such as the ELBO); and analyze the behavior of stochastic processes using tools such as eigenvalue decompositions and coupling arguments. Some questions will ask students to design or adapt algorithms in pseudocode, bridging theory and practice. The exams assess depth of understanding of the theorems, derivations, and algorithmic frameworks introduced in lecture and the textbook. **Exam 1 covers material taught from August 25 through September 24:** probability and linear algebra foundations, maps and changes of variables, ODEs and probability conservation, continuous flow matching, SDEs, Fokker-Planck, continuous diffusion, score matching, DDPMs, and guidance. **Exam 2 is cumulative and covers all material taught from August 25 through November 10:** all Exam 1 topics together with categorical corruption, continuous-time discrete diffusion, block diffusion, discrete flow matching, simplex and Gumbel-softmax flows, rectification and multi-objective guidance, flow map matching, latent and discrete flow maps, posterior guidance, and expanding state spaces.

- **Group Projects (55%):** Students will be assigned to groups at the beginning of the semester based on their level of expertise, with students at similar levels grouped together so that each team works at a comparable pace. There are two projectsL

  - **Project 1, Continuous Generative Models (15%):** Teams conduct a two-modality empirical study of continuous generative modeling. They implement and compare continuous flow matching and diffusion under a matched evaluation protocol, demonstrate guided generation, and introduce and ablate a methodological innovation. The innovation may modify the probability path, coupling, objective, guidance rule, sampling procedure, or another justified component. Teams first evaluate the method on one continuous modality and then transfer the same innovation to a second modality to test whether the improvement generalizes.
  - **Project 2, Discrete Generative Models: ICML-Ready Paper (40 points; 40%):** Teams develop a novel discrete generative formulation and produce a complete paper ready for submission to the ICML main conference. The paper must establish the novelty of the proposed formulation and include relevant benchmarks, competitive baselines, controlled ablations, and thorough quantitative evaluations with multi-seed experiments. These evaluations must demonstrate improvement and superiority over the baselines for the proposed task and support the paper's central claims. Deliverables include a complete code repository and a complete paper in ICML submission-ready format, including references and an appendix. Top papers may be submitted to ICML 2027.

    **Assignment and paper outline.** Project 2 is assigned on Thursday, October 22. By Tuesday, November 10 at 8:30 am, teams must submit an ICML submission-ready paper outline containing all non-execution portions: the abstract, introduction, preliminaries, proposed formulation, results sections with planned tables but no numerical results, discussion, all references, and appendix. The outline must explain the proposed formulation and its novelty and specify the planned experiments, including benchmarks, baselines, ablations, evaluation metrics, and experimental protocols. Sections that depend on experiments should describe the planned analyses without claiming completed results.

    **Complete submission and defense.** The complete Project 2 paper and code are due on Thursday, December 3 at 8:30 am. Project 2 defenses are held on Friday, December 4.

  **Defense format.** Every project ends with a 30-minute defense per group: a 10-minute presentation of the work followed by 20 minutes of questions from the teaching team. The code and writeup are due before class (8:30 am) on the Tuesday or Thursday before the defense, which leaves that day for the team to prepare its slides and talk. Defenses are then held the following day: Wednesday, September 30 for Project 1 and Friday, December 4 for Project 2, so that the full day can be scheduled for talks.

## Practice Problems

There are no graded problem sets. We will release practice exams with full solutions throughout the semester. These are ungraded and are meant to help you prepare for the exams and build fluency with the derivations and implementations from lecture.

## Final Grade

No curve; traditional grading scheme (90-93 = A-, 94-97 = A, 97-100 = A+). Grades posted on Canvas and directly scaled to percentage.

## Tentative Schedule

| **Date** | **Class Topic** |
|---|---|
| Tue, Aug 25 | **Lecture 1: Probability Foundations**<br>Course logistics and roadmap; continuous and discrete state spaces; probability distributions; expectation and Monte Carlo estimation; joint, marginal, and conditional probability; Bayes' rule |
| Thu, Aug 27 | **Lecture 1: The Probability Simplex and Linear Algebra**<br>Probability vectors and simplex geometry; logits and softmax; vectors and matrices; linear transformations; eigenvalues, eigenvectors, diagonalization, and matrix exponentials |
| Tue, Sep 1 | **Lecture 1: Maps, Pushforwards, and Changes of Variables**<br>Maps, inverse maps, and composition; pushforward distributions; derivatives and Jacobians; determinants and local volume change; finite change of variables |
| Thu, Sep 3 | **Lecture 1: Vector Fields, Probability Conservation, and Optimization**<br>Gradients, directional derivatives, vector fields, divergence, and the continuity equation; entropy, cross-entropy, KL divergence, constrained optimization, and the complete generative-model framework |
| Tue, Sep 8 | **Lecture 2: ODEs and Continuous Probability Transport**<br>Ordinary differential equations, velocity fields, flow maps, Euler's method, numerical integration, conservation of probability, the continuity equation, instantaneous change of variables, and continuous normalizing flows |

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| Thu, Sep 10 | **Lecture 2: Flow Matching**<br>Endpoint couplings, probability paths and interpolants, conditional and marginal velocities, the flow-matching regression objective, conditional-expectation derivation, sampling, neural parametrization, and implementation<br>*Project 1 (Continuous Generative Models) Assigned* |
| Tue, Sep 15 | **Lecture 3: SDEs, Fokker-Planck, and Score Matching**<br>Flow-matching recap, Brownian motion, stochastic differential equations, drift and diffusion, Itô's lemma, the Fokker-Planck equation, variance-preserving diffusion, Gaussian scores, and denoising score matching |
| Thu, Sep 17 | **Lecture 3: Reverse-Time Diffusion and DDPMs**<br>Noise prediction, score-driven sampling, reverse-time SDEs, DDPM forward transitions and posteriors, the learned reverse mean, time-conditioned networks, U-Nets, and DDPM training and sampling |
| Tue, Sep 22 | **Lecture 3: Diffusion Paths and Probability-Flow ODEs**<br>Latent diffusion, AMP-Diffusion, alternative noise paths, probability-flow ODEs, deterministic sampling, and the connections among score matching, diffusion, and flow matching |
| Thu, Sep 24 | **Lecture 3: Guidance for Continuous Generative Models**<br>Conditional scores, classifier guidance, classifier-free guidance, reward guidance, conditional flow matching, multi-objective guidance, time-dependent property reliability, and guided latent flow and diffusion models |
| Tue, Sep 29 | **Lecture 4: Masked Diffusion and MDLM**<br>Discrete sequence corruption, absorbing masks, exact reverse reveal probabilities, clean-token prediction, the masked diffusion ELBO, continuous-time weighting, and MDLM training and sampling<br>*Project 1 (Continuous Generative Models) Code and Writeup due at 8:30 am* |
| Wed, Sep 30 | *Project 1 defenses (Continuous Generative Models)*<br>30-minute defense per group, 10-minute talk, 20-minute Q&A with the teaching team |
| Thu, Oct 1 | **Fall Term Break (Oct 1-4)** |

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| Tue, Oct 6 | **Lecture 4: Continuous-Time Markov Chains**<br>Jump rates and rate matrices, waiting times, the master equation, matrix exponentials and transition kernels, masking and uniform noise as rate matrices, stationary distributions, detailed balance, and time reversal |
| Thu, Oct 8 | *Exam 1*<br>Material taught from August 25 through September 29: mathematical foundations, maps and changes of variables, ODEs and probability conservation, flow matching, SDEs, continuous diffusion, score matching, DDPMs, guidance, masked diffusion, and MDLM foundations |
| Tue, Oct 13 | **Lecture 4: Reverse Rates, Guidance, and the Rate-Matching Loss**<br>Reverse rates as probability ratios, the denoiser parametrization, Doob's $h$-transform for guidance, the path KL as a sum of rate errors, the rate-matching likelihood bound, and sampling with correctors |
| Thu, Oct 15 | **Lecture 4: Discrete Diffusion Models and Guided Sequence Generation**<br>Structured and uniform corruption, score entropy, block diffusion, learned per-token rates, simplex diffusion, a full implementation for DNA, planner-guided sampling, guided peptide design, and any-length generation |
| Tue, Oct 20 | **Lecture 5: Discrete Flow Matching with Jump Rates**<br>Probability paths, jump rates as probability velocities, the master equation as a continuity equation, couplings and mixture paths, conditional and marginal velocities, the denoiser and cross-entropy training, and sampling |
| Thu, Oct 22 | **Lecture 5: Correctors, Kinetic-Optimal Paths, and Edit Flows**<br>Schedules and sample quality, backward-velocity correctors, multimodal discrete flows, kinetic-optimal paths, insertions and deletions with Edit Flows, simplex-valued states, and Dirichlet conditional paths<br>*Project 2 (Discrete Generative Models: ICML-Ready Paper) Assigned* |

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| Tue, Oct 27 | **Lecture 5: Dirichlet, Fisher-Rao, and Gumbel-Softmax Flows**<br>Dirichlet velocities, scores, and guidance, Fisher information geometry and spherical geodesics, Gumbel-max sampling, differentiable Gumbel-softmax paths, temperature schedules, and the replicator equation |
| Thu, Oct 29 | **Lecture 5: Guidance, Rectification, and Multi-Objective Refinement**<br>Guidance as an exact $h$-transform, predictor-free and Taylor-approximated guidance, multi-objective guidance, reflow and endpoint recoupling, total correlation, reward-tilted targets, and Metropolis-Hastings refinement |
| Tue, Nov 3 | **Lecture 6: Flow Maps and the Three Training Identities**<br>Finite-interval flow maps, the identity, cocycle, and tangent axioms, the Jacobian equation and Liouville's formula, the Lagrangian, Eulerian, and composition identities, and MeanFlow |
| Thu, Nov 5 | **Lecture 6: Consistency, Distillation, and Discrete Flow Maps**<br>Jacobian-vector products, shortcut and consistency models, trajectory error bounds, distillation and self-distillation, a worked implementation, the two-time denoiser on the simplex, and discrete flow maps |
| Tue, Nov 10 | **Lecture 6: Stochastic, Expanding, and Guided Flow Maps**<br>Solution maps of SDEs, strong stochastic flow maps with the noise path as an input, expansion and transport with birth events, variable-length generation, reward-tilted endpoints, value-gradient guidance, and a synthesis<br>*Project 2 (ICML-Ready Paper) Outline due at 8:30 am* |
| Thu, Nov 12 | **Lecture 7: Optimal Transport, from Monge to Sinkhorn**<br>Couplings, the Monge and Kantorovich problems, duality and complementary slackness, Wasserstein distances, Brenier's theorem, the Benamou-Brenier dynamic formulation, entropic regularization, and the Sinkhorn algorithm |
| Tue, Nov 17 | **Exam 2 Review**<br>Cumulative review with the TAs to go over practice exam and remaining questions and concerns |

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| Thu, Nov 19 | *Exam 2*<br>Cumulative coverage of all material taught from August 25 through November 10: mathematical foundations, continuous flow and diffusion models, discrete diffusion, discrete flow matching, guidance, and flow maps |
| Tue, Nov 24 | **No class** |
| Thu, Nov 26 | **Thanksgiving Break (Nov 26-29) - no class** |
| Tue, Dec 1 | **Lecture 7: Continuous Schrödinger Bridges**<br>Path-space relative entropy, the static bridge and Schrödinger potentials, Girsanov's formula, stochastic control and Hamilton-Jacobi-Bellman, Doob's $h$-transform, and iterative proportional and Markovian fitting |
| Thu, Dec 3 | **Lecture 7: Discrete Schrödinger Bridges and Course Synthesis**<br>Simulation-free score and flow matching, bridges for jump processes, discrete Doob transforms, bridge matching on graphs and categorical sequences, branched and entangled bridges, and course synthesis<br>*Project 2 (ICML-Ready Paper) Code and Paper due at 8:30 am* |
| Fri, Dec 4 | *Project 2 defenses (ICML-Ready Paper)*<br>30-minute defense per group, 10-minute talk, 20-minute Q&A with the teaching team |

## Student Success

If you are having trouble contributing to or completing assignments or projects, please reach out to me so we can discuss appropriate strategies. You may email me or come to office hours to describe any personal or academic difficulties you are facing. I may also direct you to other resources on campus that can help.

The Weingarten Center for Student Accessibility provides free academic support services for all Penn students, including learning consultations, academic coaching, tutoring, and study skills resources. The Weingarten Center also coordinates academic accommodations for students with disabilities (see below).

If you are concerned about your physical or mental health, Counseling and Psychological Services (CAPS) offers individual and group counseling, workshops, and crisis support. For urgent concerns, CAPS provides 24/7 phone support at 215-898-7021. Additional wellness resources can be found through Student Health and Counseling.

## Academic Accommodations

If you need to request accommodations for a disability or medical condition, contact the Weingarten Center for Student Accessibility as early as possible. I will work with their office to ensure equal access to course materials and assessments.
Absences will be accommodated with proper documentation:

- Incapacitating illness or injury (Incapacitation Form required)
- Personal emergencies (dean's excuse required)
- Religious observances (must clear in advance)
- University athletic obligations (NOVAP required)
- Required court or legal appearances (must clear in advance)

## Academic Integrity

As members of the Penn community, students are expected to uphold the principles outlined in the Code of Academic Integrity. This code reflects Penn's commitment to scholarship, honesty, fairness, respect, and responsibility. Students must:

- Not engage in cheating, plagiarism, or other forms of academic dishonesty
- Conduct themselves honorably in all academic and nonacademic endeavors
- Seek clarification when unsure about the expectations for an assignment, exam, or group work

Please speak with me to get any clarification about this course.